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Solar radio bursts observations from Solar Orbiter — Antonio Vecchio

Vecchio, A., Maksimovic, M., Bonnin, X., Pesini, A., Kretzschmar, M., Thepthong, P.

Radboud University and Lira- Observatoire de Paris

The Radio and Plasma Waves (RPW) instrument on the ESA Solar Orbiter mission is designed to measure electric and magnetic fields, plasma wave spectra and polarization properties, the spacecraft floating potential, and solar radio emissions in the interplanetary medium. In particular, RPW observes solar radio emissions up to 16 MHz, as well as associated Langmuir waves near the local plasma frequency.

In the radio domain sampled by RPW, the most commonly observed features are solar Type III bursts. The high time and spectral resolution of RPW, combined with its unique capability to measure magnetic fluctuations up to ~1 MHz with high sensitivity, enable unprecedented sampling of Type III radio bursts from space, allowing, for the first time, the measurement of their magnetic component.

In this contribution, I review the most relevant results obtained with RPW in the analysis of solar Type III bursts, including:

- the first direct measurements of the magnetic component of solar radio emissions,
- high temporal resolution observations of Type III radio bursts in the 3-13 MHz frequency range,
- the first catalogue of interplanetary Type III radio bursts compiled through a citizen science campaign.

The Role of Particle Scattering Processes in Solar Flares — Gordon Emslie

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WKU

Emissions in radio and hard X-rays are prompt diagnostics of the high energy electrons that are accelerated in flares, which in turn provide crucial information on the primary energy release mechanism(s). For spatially resolved radiation that is optically thin, the properties of the radiation field (intensity, photon energy/wavelength/frequency, direction, and polarization), combined with the appropriate “cross-section” for the emissions being observed, provide information on the local electron phase-space distribution function. Spatially-resolved measurements can thus be used to test various electron transport models, while for spatially unresolved measurements, an appropriate electron transport model is required in order to relate the observed, target-averaged, properties of the electron population to those of the all-important accelerated population.

Models of electron transport range from the early cold-target, single particle, analytic approaches to full Fokker-Planck treatments that include physical processes such as Coulomb collisions, return current Ohmic effects, and scattering in the target. Scattering processes include both inelastic (e.g., collisional, ohmic) and elastic (e.g., magnetic, turbulent) scattering. We will review the evolution of scattering treatments over the years, and we will emphasize that even if the electron energy loss rate can be adequately modeled as a cold thick-target, with negligible diffusion in particle energy, the trajectory of the electrons, and so the spatial distribution of the electron spectrum, still fundamentally depends on diffusive angular scattering processes. This has substantial implications for the electron distribution function and its associated heating profile, and hence for the atmospheric response to flare energy input. Further, we will resolve the paradox that while in a converging magnetic field geometry magnetic scattering causes both positively- and negatively-charged particles to experience a magnetic mirror force toward the region of weaker field, there is no associated field-aligned body force that should be added to the MHD equations.

Simulations of plasma emission in plasmas with density fluctuations — KRAFFT Catherine

Krafft, C., Polanco-Rodriguez F.J., Volokitin A.S., and Savoini P.

Paris Saclay University

Plasma emission in the solar wind is profoundly influenced by random density fluctuations, which fundamentally alter the dynamics of electrostatic wave turbulence excited by electron beams during type III solar radio bursts. As these waves propagate through inhomogeneous plasma, they undergo a suite of linear transformations—including reflection, refraction, tunneling, trapping, and mode conversion into electromagnetic radiation. These interactions critically reshape the pathways to electromagnetic emission at the plasma frequency. Strikingly, the canonical nonlinear three-wave decay process, wherein a Langmuir wave decays into an electromagnetic wave and an ion acoustic wave—a mechanism long considered central to plasma emission—becomes subordinate in the presence of strong density turbulence. Instead, linear mode conversion at constant frequency emerges as the dominant process. This presentation advances our understanding of solar wind plasma emission by elucidating the pivotal role of density fluctuations in governing wave dynamics and radiation mechanisms.

K-band solar monitoring with INAF single-dish facilities: from SUNPIT to Deep Learning applications — Marco Marongiu

Marongiu M., Pellizzoni A., Zucca P., Mancini M., Righini S., Mulas S., Iacolina M.N.

INAF - Cagliari Astronomical Observatory

The solar radio emission in the K-band range (18–26 GHz) is a fundamental workbench for studying the solar chromosphere and corona. While the signal is dominated by the quiet-Sun free-free emission in local thermal equilibrium, dynamic structures - such as Active Regions, Coronal Holes, and Polar Brightenings - reveal complex interactions between solar plasma and local magnetic fields through additional gyro-magnetic components. Since 2018, the SunDish project has mapped these phenomena, generating a unique dataset of approximately 600 radio maps.

To better manage the scientific exploitation of the data, we developed SUNPIT (SUNdish Pipeline Tool). This tool was specifically designed to streamline the transition from raw data to high-level scientific products. SUNPIT optimises imaging, flux calibration, and the extraction of physical parameters, including brightness temperatures, spectral indices, and fractional polarisation. A key advancement is the integration of Deep Learning techniques, specifically Convolutional Autoencoders, to improve the identification and tracking of dynamic solar features.

In this talk, we demonstrate how this synergy between Deep Learning and high-frequency radio observations is essential for advancing our physical understanding of the Sun. This reliable and automated framework enhances the accuracy and efficiency of key investigations, including the analysis of the solar size and the solar atmosphere in terms of plasma density and temperature, the characterisation of gyro-resonance components in active regions, and the analysis of coronal holes. These tools provide the robust diagnostic framework necessary to fully exploit the INAF single-dish network for long-term solar monitoring.

Escape of energetic particles from the solar corona into the heliosphere — Mykola Gordovskyy

Gordovskyy, M., Bate, W., Brun, A.S., Sieyra, M.V., Strugarek, A. and Browning, P.K.

University of Hertfordshire

High energy ions and electrons play a crucial role in the fast energy transport in solar flares. A fraction of the particles accelerated in flares precipitate in the solar atmosphere, heating the corona and chromosphere, and producing non-thermal gamma-, X-ray and radio emissions. Others escape from the corona into the heliosphere. The escaping energetic particles are an important component of the space weather. Recent advances in computational and observational studies of solar energetic particles, have significantly improved our understanding of how particles are transported in the solar corona, and from the corona into the heliosphere, particularly thanks to the unprecedented data from Solar Orbiter and Parker Solar Probe missions. In this talk, I will review the current progress in studies of solar particle acceleration and escape into the heliosphere, with an emphasis on data-constrained computational modelling. I will present our latest results of magnetic reconnection and particle acceleration modelling in individual solar flares, focusing, in particular, on the effect of magnetic configuration and the location of the acceleration site on the properties of escaping energetic particles.

Bursty acceleration and 3D trajectories of electrons in a solar flare — Shilpi Bhunia

Hayes, A. L., Klein, L., Vilmer, N., Maloney, A. S., Gallagher, T. P.

Observatoire de Paris

Solar flares are among the most energetic phenomena in the heliosphere, emitting radiation across the electromagnetic spectrum. During flare, electrons are accelerated to non-thermal energies as a result of magnetic reconnection. These electrons then propagate upwards and downwards from the energy release site along magnetic field lines and produce radio and hard X-ray (HXR) emissions, respectively. Frequently, we observe the simultaneous occurrence of multiple HXR peaks and successive radio bursts, providing compelling evidence that these emissions originate from a common population of accelerated electrons and that the underlying energy-release process consists of multiple local acceleration process. We present a detailed analysis of a highly energetic flare observed by the Spectrometer Telescope for Imaging X-rays (STIX) onboard Solar Orbiter, in conjunction with coordinated radio observations from ground- and space-based instruments from the low corona to interplanetary space. By investigating the evolution of X-ray, and radio sources, we find that the electrons producing HXR and radio emission have similar acceleration origins. Importantly, our study provides observational evidence that the flare acceleration process is temporally and spatially fragmentary, and during each of these small-scale processes, the electron beams are injected into a very fibrous corona and produce complex HXR and radio emission

Radio Observations of Solar Active Regions with EOVSa — Sijie Yu

Sijie Yu

New Jersey Institute of Technology

Solar active regions are important radio sources that provide diagnostics of coronal plasma and magnetic fields. This talk will review active-region radio observations with an emphasis on recent results from the Expanded Owens Valley Solar Array (EOVSA). I will briefly describe the current EOVSa imaging spectroscopy pipeline and its capabilities for routine multi-frequency studies. I will then discuss the main emission mechanisms in active regions, including gyroresonance, gyrosynchrotron, and free-free emission, and highlight key applications such as coronal magnetic field measurements.

ALMA Observations of Solar Flares — Stephen White

White, S.

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Solar flares have not been a primary target for ALMA partly due to uncertainty about the response to bright flare emission at millimeter wavelengths. However, in 2024 May during the first ALMA observations where circular polarization measurements of the Sun were permitted, several flares were successfully observed at arcsecond resolution, including an X flare. Since the quiet Sun flux in the ALMA primary beam at 100 GHz is around 60 sfu, fluxes of that order or less can readily be handled by ALMA. On behalf of the team involved, this paper will discuss the observations in the context of models of flare heating and the response of the chromosphere to electron beams, and directions where ALMA can help with flare studies.

First imaging observations of the evolution of Coronal Mass Ejections and Eruptive Prominences at K-band (18-26 GHz) — Alberto Pellizzoni

Pellizzoni A., Amaro A., Napolitano D., Mulas S., Marongiu M., Egron E., Righini S., Sirigu M. on behalf of the SunDish Team

INAF - Italy

The radio emergence of coronal mass ejections (CMEs) and eruptive prominences (EPs) has been the subject of several studies, given their relevant role in exploring the solar atmosphere and modelling its magnetic restructuring processes. This modelling is key to understanding the energy release mechanisms of active regions (ARs), which are among the main drivers of space weather.

In this work, we present and analyse an unusual detection of an EP/CME event at 18.8 and 24.7 GHz performed by the Sardinia Radio Telescope (SRT) on August 19 and 20, 2025, under the SunDish project (<https://sites.google.com/inaf.it/sundish>).

We discuss the complex structure of the erupting event, including an unprecedented radio feature extending over 15 arcmin from the solar disk and emerging from a region characterised by reconnection threads involving helical kink instabilities, ARs, and filaments. Through spectral and polarimetric information, we can constrain EP/CME physical parameters, including an extrapolation of the magnetic field associated with gyrosynchrotron and synchrotron emission components.

This work paves the way for the systematic search for CME events in high-frequency radio images and their correlation with standard coronagraphic images and space-based measurements.

Transient brightenings around a stable sunspot detected by ALMA at 3 mm — Alexander Nindos

Nindos, A., White, S.M., Alissandrakis, C.E., Barta, M., Bastian T.S., Berlicki, A., Brajsa, R., Chen, B., De Pontieu, B., Eklund, H., Fleishman, G.D., Gary, D.E., Hudson, H.H., Iwai, K., Kobelski, A., Krucker, S., Loukicheva, M., Reardon, K., Sanchez-Maes, S., Shimojo, M., da Silva Santos, J., Skokic, I., Solanki, S., Szydlarski, M., Wedemeyer, S.

University of Ioannina, Greece

Using Atacama Large Millimeter/submillimeter Array (ALMA) observations, we performed the first systematic survey for weak small-scale transient brightenings around a stable sunspot at 3 mm. Our dataset included images of an 80" x 80" field of view obtained with angular resolution of a few arcsec at a cadence of 2 s. The transient brightenings were detected as weak enhancements above the average intensity. Many of them were also registered in Atmospheric Imaging Assembly (AIA) images obtained at several EUV channels. All detected events were thermal in nature and the strongest of them showed loop-like or jet morphologies. Their properties were checked against model calculations and MHD simulations of jets.

Using CoSEE-Cat to investigate the relation of solar energetic electron events with X-ray flares and type III radio bursts — Alexander Warmuth

Warmuth, A.

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We have developed the Comprehensive Solar Energetic Electron event Catalogue (CoSEE-Cat), a new multi-instrument resource combining in situ and remote-sensing observations from Solar Orbiter to systematically characterise solar energetic electron (SEE) events and their solar origins. The first data release covers 303 SEE events detected between November 2020 and December 2022 at heliocentric distances ranging from 0.3 to 1.0 au, compiled through a joint effort of eight instrument teams (EPD, STIX, EUI, RPW, Metis, SoloHI, SWA, and MAG). Statistical analysis of this first data release reveals a clear dichotomy between impulsive and gradual SEE events, supporting flares and CMEs as the respective sources for these two populations. We further show that commonly reported delays between electron solar release times and associated X-ray and type III radio burst signatures grow systematically with heliocentric distance, indicating that interplanetary propagation effects rather than actually delayed injections play a significant role.

Location and properties of fine frequency structures in the spectra of solar type IV radio bursts — Aneesh Nadiger

Nadiger A, Morosan D E

University of Turku

ABSTRACT

Context : The Type IV continuum spectra is dynamic and many short-lived fine structures have been observed. Several observations of stripes of emission and absorption bands forming zebra-like or fiber-like structures against the background of a continuous radio spectra have been reported.

Aim : In this study, we investigate zebra patterns. Our analysis focuses on identifying the sources of the Zebra patterns and determining their properties.

Methods : We study the Zebra bursts observed on 28th March, 2022. We used spectroscopic data from the e-Callisto network to identify the Zebra patterns and their characteristics, radio imaging from the Nançay Radioheliograph to locate the emission sources, and ultraviolet imaging from AIA to provide coronal imaging.

Results : We observe two distinct regions of Zebra patterns in our study. The sources are found to be stationary, with a period of about 6 seconds and drift rates of 7–9 MHz s⁻¹. The instantaneous separation between the stripes is about 4 MHz. Comparison with previous observations shows that Δf typically increases with increasing frequency.

Conclusions : The confined location of the Zebra sources would imply to the sources are trapped, possibly in a coronal loop. We observe 6-8 second periods for the zebras, indicating that the trapped plasma undergoes fast sausage mode oscillations

Solar Radio Type V Burst: Scattering, Trapping, and Emission Process of a Dense Electron Beam — Arnold O. Benz

Benz, A.O., Huber, C., Timmel, V., Monstein, C.

FHNW Windisch and ETH Zurich

Solar type V radio bursts are associated with type III bursts. Several processes have been proposed to interpret the association, the electron velocity distribution, and the emission. We present the observation of a unique type V event observed by e-CALLISTO on 2121/05/07. The type V radio emission follows a group of U bursts. Its polarisation is opposite to the type III burst and its front edge has a U-shape. The observations conform to a scenario where a dense electron beam first saturates its two-stream instability (causing type III emission) and then becomes unstable to another instability, such as the electron firehose instability. This second instability scatters some beam electrons into velocities perpendicular to the magnetic field and produces a trapped distribution prone to electron cyclotron masering (ECM). A reduction in beaming is predicted for electron beams continuing to interplanetary space, observable for Parker Probe and Solar Orbiter.

Dynamics of type II solar radio burst sources and spectral bridge formation — Artem Koval

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Type II solar radio bursts are commonly associated with shock waves propagating through the solar corona in connection with coronal mass ejections. A characteristic feature of these bursts is band splitting, where the emission lane divides into two closely spaced, nearly parallel bands. While its origin is still debated, it is often interpreted in terms of either plasma emission from different shock regions (upstream and downstream areas) or spatially separated acceleration sites along the shock front. We analyze LOFAR observations of a type II radio burst that display fine spectral structures connecting the split bands, referred to here as spectral bridges. These features exhibit both positive and negative frequency drifts. Their properties suggest that they are not random appearances, but are instead linked to evolving physical conditions within the shock or/and the coronal environments. In this study, we explore a scenario of the formation of spectral bridges. In addition, we examine the motion of the emission sources associated with the split bands and discuss how their dynamics may reflect the underlying structure and evolution of the shock.

Characteristics of Pulsating Structures Obtained from ARTEMIS IV/JLS and NRH — Bouratzis Costas

Bouratzis, C., Alissandrakis, C. E., Hillaris, A., Armatas S.

University of Athens, Greece

We analyzed the characteristics of pulsating structures embedded in type IV radio-bursts, observed by the ARTEMIS-IV/JLS radio spectrograph and recorded with the SAO high resolution (100 samples per second) receiver in the 270–450 MHz range, with a frequency resolution of 1.4 MHz. We present preliminary results for their frequency drift, duration and periodicity, using cross correlation, autocorrelation, and a maximum tracking method. Imaging information was obtained from the Nançay Radioheliograph.

The effect of solar wind turbulence on the clumpiness and local generation of Langmuir waves — Camille Yasmina Lorfing

Lorfing, C.Y., Reid, H., Pisa, D., Nicolaou, G., Trotta, D., Krupar, V., Magdalenic, J.

KU Leuven

How do solar wind turbulence affect the clumpiness and local generation of Langmuir waves? Solar electrons beams are accelerated in the corona to then travel out into the solar wind. These beams of non-thermal electrons evolve as a function of distance from the Sun, interacting with the turbulent background plasma and growing Langmuir waves as they propagate. Subsequent radio emission is also seen in the form of type III bursts. Where solar wind turbulence comes from and how turbulent energy is dissipated and evolves within the heliosphere has been a long lasting science question, studied through numerous simulations and observations since the 1970s. However, previous studies looking at the role of turbulence in the clumpiness and generation of Langmuir waves have never done so at such a high temporal resolution as offered by Solar Orbiter. Through a combination of in-situ and remote sensing observations of plasma conditions at the spacecraft and associated radio emission, we study the influence of density fluctuations in the solar wind on the local generation of plasma waves between 0.3 and 1 AU. With 35 events between November 2020 and January 2025, we produce a statistical study of the role of turbulence on wave-particle interactions with heliocentric distance, and different latitudes outside of the ecliptic plane. This allows us to determine a relationship between density fluctuations, solar wind speed, heliocentric distance, and the clumpiness and waveform of Langmuir waves. We also study the associated radio emission to indentify a link between the intensity and maximum frequency of this radio emission and the quantity of Langmuir waves growth above thermal level at the spacecraft. Understanding the mechanisms driving Langmuir waves growth in different solar wind conditions will permit more accurate predictions of wave-particle interactions and propagation in the heliosphere. Moreover, our work not only supports previous theoretical work but can be complemented by measurements from the Parker Solar Probe spacecraft.

Evolution of coronal radio wave scattering during an M class solar flare — Christian Vocks

Vocks, C., Bisi, M., Dabrowski, B., Morosan, D., Gallagher, P., Krankowski, A., Magdalenic, J., Mann, G., Marque, C., Matyjasiak, B., Rothkaehl, H., Zucca, P.

Leibniz-Institut für Astrophysik Potsdam (AIP)

We present LOFAR observations of an M class flare with intense type III radio bursts. Some isolated burst have a fundamental-harmonic structure, but for most bursts this is not visible due to a rapid succession of bursts. Spectroscopic imaging with LOFAR shows type III bursts as compact sources for a given frequency. The intensity varies with burst evolution, and transient dual-source structures appear. We interpret these as signatures of fundamental and harmonic emission, the latter from a higher location in the corona. Fundamental-harmonic pairs, e.g. fundamental emission at 35 MHz and harmonic emission at 70 MHz, should originate from the same plasma volume. Differences in their positions and intensity variations are expected since radio wave transport effects in the corona, like scattering and refraction, affect fundamental more than harmonic emission. We show the evolution of position differences of fundamental-harmonic pairs during different phases of the flare. Since the effect of refraction in the overall density structure of the corona is not expected to change much, this enables a study how the flare influences coronal radio wave scattering.

The Small Phased Array DEMonstrator (SPADE) - Description and first results — Christophe Marqué

Marqué, C., Martínez Picar, A., Magdalenic, J., Tassan-Din, E.

Royal Observatory of Belgium

Since the summer 2024, a compact phased array made of eight identical antennas is monitoring solar activity in the decameter range at the Humaïn radioastronomy station in Belgium. The instrument relies on Software Defined Radio receivers and open source software libraries (GNU Radio) for operations related to pointing and production of dynamic spectra between 25 and 75 MHz.

We describe here the main technical aspects of SPADE and present highlights of observations performed since the instrument was turned on.

Spicule modeling from simultaneous ALMA and IRIS observations — Costas Alissandrakis

Alissandrakis, C.E., Vial, J.-C. and Bastian T. S.

University of Ioannina, Greece

We analyze observations of spicules in a coronal hole with ALMA at mm- λ and IRIS spectra in the Mg II k and h lines. We computed the mm emission as thermal free-free and the Mg II emission through a one-dimensional, static, isothermal and isobaric NLTE model with a vertical slab geometry. We found that the electron temperature increases smoothly from about 10000 K at a height of 2000 km to about 17000 K at 14000 km and that the electron density decreases from about 10^{11} cm⁻³ to 10^{10} cm⁻³ in the same interval. We provide good evidence that an important part of the intensity decrease at the center of the k and h lines is due to opaque foreground plasma. we verified that the turbulent velocity in spicules is very high, up to 25-30 km/s, which could be an indirect signature of shocks.

Magnetic Field Geometry and Anisotropic Scattering Effects on Solar Radio Burst Observations — Daniel Clarkson

Clarkson, D., Kontar, E.

University of Glasgow

Solar radio bursts and noise storms show complex fine structures whose observed properties are strongly affected by propagation through the turbulent corona. We investigate how anisotropic radio-wave scattering combined with non-radial magnetic field geometry shapes the apparent motion and size of fine structures. Using radio-wave propagation simulations with the dipolar structure of an active region, we reproduce puzzling non-radial source motion observed by LOFAR parallel to the solar limb. The apparent source evolution follows the direction of magnetic field lines rather than density gradients, and the major axis aligns perpendicular to the local magnetic field. Strong density fluctuation anisotropy within non-radial fields can also lead to source bifurcation, producing multiple apparent components. We also show that anisotropic scattering within closed magnetic structures can produce compact apparent sources, consistent with LOFAR observations of type I noise storms. Despite their diverse spectral signatures, noise storm embedded fine structures (type I bursts, S-bursts, spikes) exhibit similar compact sizes, which we show arise naturally from emission within closed fields. Secondary effects include enhanced coronal densities, steeper density gradients, reduced turbulence levels, and strong fluctuation anisotropy. These results demonstrate that the imaged source locations and sizes of radio bursts are governed primarily by propagation effects in combination with the large-scale magnetic topology rather than intrinsic emission differences, highlighting the crucial role of magnetic structure in shaping and interpreting scattered solar radio burst observations.

From Global Networks to High-Resolution Arrays: Unified Observations of Solar Radio Bursts — Daniela Banyś

Banyś, D., Heinrich, L., Tandler, F., Wenzel, D., Monstein, C., Dąbrowski, B., Krankowski, A., Vocks, C.

German Aerospace Center (DLR e.V.)

Ground-based radio networks are crucial for the continuous monitoring of solar eruptive events and their impact on technological systems, although maintaining a reliable and globally distributed network remains challenging. We present a comparative study of solar radio burst observations and their relevance for space weather monitoring using CALLISTO, RSTN, and LOFAR. The CALLISTO spectrometers, operated within the international e-CALLISTO network (>190 stations worldwide), provide routine monitoring of solar radio emission across a broad frequency range from approximately 10 up to 1600 MHz. Within this framework, DLR operates optimized compact CALLISTO receivers, covering 10–80 MHz, 45–860 MHz, 800–1000 MHz, and 1.0–1.6 GHz. This system enables investigations of flare-accelerated electrons, shock-related emission, and coronal mass ejection signatures, while ensuring good comparability through standardized instrumentation.

The Radio Solar Telescope Network (RSTN) provides complementary solar radio observations from 25–180 MHz, with near-real-time event reporting via NOAA/SWPC. LOFAR, operating within 10–90 MHz and 110–240 MHz, offers exceptional sensitivity and high frequency and temporal resolution, enabling detailed studies of fine structures in solar radio bursts and early coronal emission processes.

In this work, we analyse selected solar radio events and perform direct comparisons to assess sensitivity, spectral resolution effects, and the capability to detect and characterize different solar emissions. In addition, a filtering technique is proposed that harmonizes frequency coverage across different spectra, suppresses noise, and reduces radio-frequency interferences (RFIs), thereby improving data quality for automated burst detection and classification. Our results highlight the complementary capabilities of the different networks and furthermore demonstrate the potential of combining multi-instrument radio spectra to improve space weather monitoring.

Towards daily high-frequency radio imaging of the Solar disk with the SunDish project — Davide Napolitano

Napolitano, D., Pellizzoni, A., Righini, S., Marongiu M., Mulas, S., on behalf of the SunDish team

INAF-IRA and University of Bologna

In the last 8 years, under the Sundish Project (Pellizzoni et al. 2022), the Italian network of single dish radiotelescopes, operated by the National Institute for Astrophysics (INAF), has been adapted to perform solar observations, and has been used to conduct double frequency solar imaging in the radio K-band, at 18 and 26 GHz, with a resolution ranging between 0.6 and 1 arcmin. Solar radio imaging observations in the K-band represent a unique tool, specifically probing the solar chromosphere and the transition region, where energetic phenomena originate, such as flares and Coronal Mass Ejections (CMEs). Through On-the-Fly radio imaging in this domain, it is possible to monitor the physical evolution of Active Regions (ARs) and Coronal Holes (CHs) before and after flare/CME occurrences, based on spectral index measurements and polarimetric imaging. This provides both physical insights about ARs temperature, density and magnetic field evolution during energetic events, and Space Weather applications.

We present our updated observing strategies and performances when conducting daily monitoring of the Sun at high radio frequencies, improving the current weekly schedule.

Specifically, we present a first run of daily observations for a continuous 14-day period to be performed in May 2026 with the 32-m “Grueff” radiotelescope, with a specific focus on the study of ARs and their short-term evolution. With the daily monitoring of a significant sample of flaring and non-flaring ARs, improvements can be made on the previous more sporadic diachronic observations. In addition to daily images of the full solar disk at 18 and 26 GHz, this observational run will test methods to track specific ARs and record their flux over short time-scales (ms time resolution) through high-frequency dynamical spectral analysis. We also plan to complement the above run with simultaneous observations with other ground-based and space-based observatories, in order to obtain an unprecedented probe of the dynamics of the entire Solar chromosphere. In perspective, the project aims to make use of the radiotelescope in Noto (Sicily), soon to be made available for solar observations, to improve the time coverage of our observations. Additionally, the planned implementation of tri-band receivers (K, Q, W Bands) at the INAF facilities will provide unprecedented radio coverage up to 100 GHz, enabling simultaneous observations at multiple frequencies, and allowing for the determination of the real-time spectral index of solar features.

Probing the Solar Corona with MeerKAT: Early Results and Diagnostic Capabilities — Deepan Patra

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National Centre for Radio Astrophysics, Tata Institute of Fundamental Research, India

Understanding the dynamics and structure of the coronal plasma and magnetic field requires a multi-wavelength and multi-instrument approach. Radio observations offer an independent and complementary diagnostic of coronal density, magnetic field, and non-thermal electron populations and their transportation. MeerKAT solar observations in the 580-1700 MHz provide spectroscopic snapshot imaging of the solar corona, at high temporal and spectral resolutions, probing low coronal heights (~ 1.01 – 1.2 solar radii). Its dense uv-coverage, wide field of view, and high sensitivity enable high fidelity full-polarization imaging of multiscale structures in both quiescent and active corona. These observations can probe multiple emission processes, including thermal bremsstrahlung, gyrosynchrotron, and gyroresonance emission, with polarization measurements providing additional constraints on magnetic field strength and topology. This complements EUV observations by probing coronal plasma and magnetic fields even in temperature regimes where EUV diagnostics become limited or unavailable. We present early results from MeerKAT solar observations, including imaging of coronal holes, filaments, cavities, and eruptive events. Full-Stokes imaging reveals complex polarization signatures, including circular polarization inversion above an active region across frequency along with temporal evolution during eruptive events, likely associated with mode coupling and changes in coronal magnetic topology, respectively. These results highlight the potential of MeerKAT to provide new constraints on coronal magnetic field structures and plasma conditions with unprecedented detail, particularly when complemented with multiwavelength observations. They further lay the groundwork for future solar science with the SKA-Mid telescope.

First Remote Sensing Measurement of CME-entrained Magnetic Field in the Inner Heliosphere — Devojyoti Kansabanik

Devojyoti Kansabanik, Angelos Vourlidas, John Morgan, Surajit Mondal, Divya Oberoi

Institute of Astrophysics of Andalusia, Granada, Spain

Determining the magnetic field of coronal mass ejections (CMEs) in the inner heliosphere remains both a critical and a challenging problem in heliophysics. In the past, Faraday rotation (FR) measurements of background linearly polarized radio sources occulted by CMEs have been used to estimate CME-embedded magnetic fields, but such studies have been largely confined to the outer corona (~ 20 solar radii), mostly using the Jansky Very Large Array. Extending these measurements into the inner heliosphere is significantly more demanding, as both the electron density and magnetic field strength of CME plasma decrease with distance, resulting in a much weaker FR signal. Addressing this challenge requires observations at meterwave range along with instruments that provide a wide field of view to track the expanded CME structure. The Murchison Widefield Array (MWA) fulfills these requirements. However, estimating CME-associated FR also necessitates the use of non-trivial techniques to effectively isolate and remove spatio-temporal ionospheric FR variations from the measured FR. In this presentation, we report the first successful measurements of FR from a handful of CMEs observed with the MWA in the inner heliosphere, extending out to ~ 60 solar radii. We also discuss how future observations with SKA-Low will significantly enhance these capabilities, enabling more robust modeling of CME magnetic fields constrained by heliospheric observations.

Determining the possible acceleration regions of in situ electrons using remote radio imaging and X-ray observations — Diana Morosan

Morosan, D. E.

University of Turku

Energetic particles in the heliosphere are produced by flaring processes on the Sun or shocks driven by coronal mass ejections. These particles can be detected remotely through the electromagnetic radiation they generate (X-rays or radio emission) or in situ by spacecraft monitoring the Sun and the heliosphere. Here, we investigate the acceleration locations, escape, and propagation directions of electron beams producing radio bursts observed with the Low Frequency Array (LOFAR), Nançay Radioheliograph (NRH) and NenuFAR and compare it to hard X-ray (HXR) emission and in situ electrons observed at spacecraft. These observations are combined with three-dimensional (3D) renderings of the electron acceleration locations and results from magneto-hydrodynamic (MHD) models of the solar corona in order to investigate the origin and connectivity to spacecraft. This is done to spatially relate the electrons observed remotely at the Sun to in situ electrons. In the events studied, not all radio bursts or hard X-ray locations have a direct connectivity to spacecraft observing SEPs, however, there are occasions where the connecting field lines intersect the 3D locations of type II or herringbone bursts. In addition, for these events we found comparable start times of the radio emissions at the Sun to the injection times of spacecraft electrons.

Chasing Solar and Heliospheric Activity on cue with SKAO Pathfinders and Precursors — Divya Oberoi

Oberoi, D. (1), Patra, D. (1), Kansabanik, D. (2), Dey, S. (1), Vourlidas, A. (3), Morgan, J. (4), Mondal, S. (1), Sharma, R. (5) - (1) National Centre for Radio Astrophysics, Tata Institute of Fundamental Research, India; (2) Institute of Astrophysics of Andalusia, Spain; (3) Johns Hopkins University Applied Physics Laboratory, USA; (4) Commonwealth Scientific and Industrial Research Organisation, Australia; (5) Indian Institute of Technology Kanpur, India

National Centre for Radio Astrophysics

Unsurprisingly, it has been the case for quite some time that the state-of-the-art, versatile, and non-solar dedicated instruments deliver an imaging quality much better than what is feasible with the dedicated solar observing facilities. Realising the full potential of these instruments for solar science is, however, challenging. Some of the most interesting times for solar observations are during periods of enhanced solar activity. Solar activity, however, is inherently unpredictable, especially from the perspective of scheduling observations, often done months in advance. At the same time, these telescopes are highly oversubscribed, with only a limited amount of time available for solar and heliospheric observations. A practical way to dramatically improve the efficiency of the limited observing time for capturing solar activity is by observing only when it is known to be occurring via triggered observations. As a part of preparing for science with the Square Kilometre Array Observatory, we have been working on designing, implementing, and operationalising such systems at the precursors and pathfinders for the SKAO. The first such system for capturing solar radio bursts, relying on an external dynamic spectrum-based near-realtime trigger, has recently been implemented at the Murchison Widefield Array (MWA), a precursor for the SKA-Low. Another realtime system designed to schedule observations needed to capture coronal mass ejections (CMEs) as they propagate away from the Sun has been made operational at the MWA as well as ASKAP, a pathfinder for the SKA-Mid. In addition to these remarkably successful applications, the use of triggering systems is also being explored at dedicated solar observing facilities to identify the times and durations for which data should be saved to disk or saved at higher temporal/spectral resolution. We present an overview of the current status of efforts towards triggered observations of solar and heliospheric transients, along with suggestions for their implementation at SKA telescopes.

Advancing in-situ diagnostics of solar wind heating with radio observations — Eduard Kontar*Eduard P. Kontar, A. Gordon Emslie, Daniel L. Clarkson, and Alexander Pitna*

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Plasma turbulence cascading from MHD to kinetic scales in the heliospheric plasma is believed to play a key role in coronal heating and fast solar wind acceleration, but the properties of the turbulence remain poorly constrained by observations. We compare the ion-scale density fluctuation levels inferred from the properties of solar radio bursts with the magnetic field fluctuation levels obtained through in situ measurements in the inner heliosphere. We find that the observed magnetic and density fluctuation amplitudes are consistent with excitation by kinetic Alfvén waves (KAWs) and/or KAW structures over a broad range of distances from the Sun. We then use the radio diagnostics and the KAW scenario to deduce the radial variation of magnetic fluctuation amplitudes in regions close to the Sun where in situ measurements cannot be obtained. Further, we calculate the energy cascade rate (plasma heating rate) profile over a region that extends from the low corona into the heliosphere, and compare it to the energy deposition rate required to drive the solar wind. The results provide unique diagnostics of the ion-scale plasma turbulence amplitude and energy cascade rate spanning over 3 orders of magnitude in solar distance.

From GX Simulator to pyGX: Data-Constrained 3D Modeling of Solar Active Regions with Multi-View EUV and Microwave Observations — Gelu M. Nita

Gelu M. Nita, Sijie Yu, Alexey A. Kuznetsov, Viktor Fedenev, Alexey G. Stupishin, Bin Chen, and Gregory D. Fleishman

New Jersey Institute of Technology

Understanding the magneto-thermal structure of solar active regions requires data-constrained three-dimensional (3D) modeling that integrates realistic magnetic and thermal structures with multi-wavelength observational diagnostics. The Coronal Heating Modeling Pipeline (CHMP), originally implemented within the GX Simulator [1] framework, provides a well-established methodology combining magnetic field extrapolations, field-aligned heating parameterizations, and forward-modeled emission, validated in multiple peer-reviewed studies [2–4]. Here we present pyCHMP, the first fully integrated, science-ready Python implementation of the CHMP framework, developed within the emerging pyGX ecosystem. The pipeline builds on two core components: pyAMPP [5], which constructs data-constrained magnetic models of active regions, and pyGXrender [6], which dresses the magnetic skeleton with a parameterized thermal structure and synthesizes multi-wavelength emission from Earth and spacecraft perspectives. The transition from the IDL-based GX Simulator to a Python framework is carried out within the SUNCAST (Software UNified Collaboration for Advancing Solar Tomography) platform and aligned with the evolving SunPy ecosystem, ensuring long-term maintainability and broader community accessibility. We demonstrate pyCHMP using an active region observed simultaneously by EOVS, SDO, STEREO-A, and Solar Orbiter. By iteratively adjusting the free parameters of the EBTEL-based thermal structure, we obtain model realizations that reproduce both EUV morphology and microwave emission, significantly reducing ambiguities inherent in single-instrument analyses. This enables quantitative assessment of coronal heating and its relationship to the underlying magnetic structure and establishes pyCHMP as a key step toward a modular, open, and reproducible framework for data-constrained solar modeling. Future work will extend this approach toward flare modeling within the broader pyGX ecosystem.

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Highlights of the microwave imaging spectroscopy of solar flares — Gregory Fleishman

Fleishman, G. D., Kaltman, T, Gary, D. E., Chen, B, Yu, S., Oparin, I., and Nita, G. M.

NJIT-CSTR/KIS

Over recent decade, a novel methodology that employs microwave imaging spectroscopy data opened an entirely new window to look at coronal phenomena; notably solar flares. This methodology yields evolving maps of such coronal parameters at the flare sites as the magnetic field and the thermal plasma density along with parameters of the nonthermal electrons, primarily, their number density and spectral index in their distribution over energy. These diagnostics is the key to uncover otherwise elusive physics of magnetic energy transformations and particle acceleration in solar flares. Indeed, release of the magnetic energy due to reconnection is believed to drive such transient phenomena as solar flares, eruptions, and jets. This energy release should be associated with a decrease of the coronal magnetic field. Therefore, the quantitative measurements of the evolving magnetic field strength in the corona, that we are reporting here, are required to find out where exactly and with what rate this decrease takes place. Here, we report microwave observations of several solar flares, showing spatial variations and temporal changes in the coronal magnetic field at various coronal locations [1-4]. We discovered a fast decay of the magnetic field in the coronal loops and cusp regions [1,3]; well below the nominal reconnection X point implied by the standard model of solar flare. The field decays at a rate ranging from a fraction of Gauss up to ~ 10 Gauss per second in various events and locations. This fast rate of decay implies a highly enhanced, turbulent magnetic diffusivity and sufficiently strong electric field to account for the particle acceleration that produces the microwave emission. In addition, we report such phenomena, that are beyond the standard model, as bulk electron acceleration in the cusp region [5], where the magnetic field displays the fast decay, and a new high-energy, MeV-peaked, electron component in a corona volume [6]. We report 3D coronal maps of such important parameters as Alfvén speed and plasma beta [4]. Finally, we review novel results on the spectral energy evolution of the nonthermal electrons accelerated in flares, oscillations of physical parameters in event(s) with quasiperiodic pulsations, and signatures of anisotropic pitch-angle distributions.

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Multi-wavelength Analysis of Relativistic Electron Dynamics in the X8.2 Limb Flare SOL2017-09-10 —

Guillermo Giménez de Castro

Perriyil, S. M., Giménez de Castro, G., Simões, P.J.A., Chen, B., Yu, S.

Centro de Rádio Astronomia e Astrofísica Mackenzie

The X8.2 solar flare of 10 September 2017 (SOL2017-09-10) was one of the most powerful events of Solar Cycle 24, offering a unique opportunity for multi-wavelength diagnostic studies near the solar limb. In this work, we present a comprehensive multispectral analysis of the flare's relativistic electron populations, utilizing microwave data from the Extended Owens Valley Solar Array (EOVSA), sub-millimeter observations from the Solar Submillimeter Telescope (SST), and X-ray to gamma-ray data from RHESSI and Fermi. Our analysis reveals two distinct non-thermal electron populations: a coronal source characterized by a mild electron power-law index and a deeper chromospheric source with a significantly harder index. These radio findings are remarkably consistent with the X-ray and gamma-ray spectral analysis. As previous works have shown, spatial comparisons between RHESSI 35 keV images and EOVSA microwave maps indicate that mildly relativistic electrons are confined within coronal magnetic loops. In contrast, the >1 MeV electrons responsible for the sub-millimeter and gamma-ray emission originate from a deeper atmospheric layer. Despite the occultation of footpoints due to the flare's location on the West limb, these findings suggest a clear decoupling between different energy populations.

Solar and Heliospheric Radio Emission with SKA-Low: Observational Opportunities and Inference Challenge

— Hariharan Krishnan

Hariharan Krishnan

SKAO

The solar corona and inner heliosphere constitute a dynamic and highly structured plasma environment that generates a wide range of low-frequency radio emissions, including type II and type III bursts, fine spectral structures, and transient polarized signals associated with particle acceleration and shock-driven processes. These emissions provide a powerful window into solar and heliospheric plasma physics, including electron acceleration, wave–particle interactions, and the evolution and propagation of large-scale disturbances through the heliosphere.

The SKA-Low Frequency Telescope (SKA-Low) will provide a transformative leap in solar and heliospheric radio observational capability. With its dense short-baseline configuration, broad instantaneous bandwidth (50–350 MHz), and high temporal, spectral, and spatial resolution, SKA-Low will enable sub-second imaging spectroscopy of impulsive solar radio emission with unprecedented fidelity. This will open a new observational regime for tracking the dynamics and heliospheric evolution of impulsive radio emission, resolving fine spectral structures, improving sensitivity to faint and diffuse emission, and enabling large-scale statistical studies across diverse solar and heliospheric conditions.

At the same time, the interpretation of low-frequency solar and heliospheric radio emission is intrinsically shaped by propagation effects, including scattering, refraction, and mode coupling. These processes do not limit the observational capability of SKA-Low, but define the framework within which physical inference must be made. Recognizing and incorporating these effects is essential for extracting robust and physically meaningful constraints on source and propagation properties.

In this contribution, I highlight the exceptional scientific potential enabled by SKA-Low for solar and heliospheric physics and emphasize the importance of a physically grounded interpretation framework to fully realize that potential. By clearly articulating what SKA-Low can directly reveal and what requires model-dependent inference, I aim to support optimal exploitation of its capabilities and strengthen the connection between observations and plasma physics across the solar–heliospheric system.

Study of Particle Signatures and CME-CME interaction during the 11–12th Nov 2025 event — Harsh Bhati

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The intense solar activity of 11-12 November 2025, featuring an X5.1 flare from sunspot region 4274 and a G4-class Coronal Mass Ejection (CME), triggered significant interplanetary disturbances and the strongest Ground Level Enhancement (GLE #77) in two decades. This study investigates particle signatures and CME-CME interactions to understand how shocks and plasma compression drive relativistic particle production. The research aims to characterize the restructuring of plasma and the resulting complex ejecta that led to a minimum Dst index of -217 nT during 11-12th Nov 2025. Multi-platform observations from Aditya-L1 (ASPEX-STEPS), ACE, WIND, and OMNI were used to analyze high-resolution IMF and plasma data at the L1 point. Interactions were identified through primary and secondary jumps in plasma beta ratio, velocity, density, and temperature during the passage of trailing CMEs. Results show the primary CME shock arrived at L1 at ~23:48 UT on 11 November, followed by secondary enhancements confirming the formation of complex ejecta, which are the optimal site for SEP acceleration. To quantify turbulence, magnetic field Power Spectral Density (PSD) was computed using Welch's method. Spectral analysis revealed a kinetic-scale slope of -2.602, which lies between KAW-like and Whistler-like slope, indicating energy transfer to dissipative scales. Neutron monitors confirmed GLE #77 with a rare double peak, highlighting multi-stage ejections. These results demonstrate that CME-driven shocks significantly enhance small-scale turbulence and plasma compression. The findings emphasize the non-linear nature of complex ejecta, necessitating multi-point observations and can be further improved by ML/DL analysis.

When, where, and how can fast protons contribute to coronal heating? — Hugh Hudson

Hugh Hudson, Lyndsay Fletcher, Graham Kerr, and Alec MacKinnon

U of Glasgow / SSL, UC Berkeley

Closed magnetic fields in the solar corona (the streamer belt and/or active regions) may trap high-energy ions, such as cosmic rays, with hours-long lifetimes against collisional losses. Populations of MeV-GeV positive ions may appear in the corona as a result of flare or CME particle acceleration, and their presence may be hard to detect via remote-sensing astronomical observations. We consider their potential consequences in various contexts, including the EUV late phase (ELP) and long-duration γ -ray (LDGRF) phenomena, as well as background galactic cosmic rays. The very simple model described here assumes the evolution of the trapping volume via a sequence of equilibrium states that include the reservoir property of closed magnetic fields linking the corona to the chromosphere. Based upon a basic static equilibrium model (RTV), we estimate the particle numbers and energies for which collisional heating rates match nominal coronal heating. We find that a population of MeV protons at a number density of order 10^{-6} of the background plasma density can produce significant coronal heating on ELP/LDGRF time scales of hours. We comment also on improving the RTV-based analysis by computing RADYN equilibria with particle energy input balanced by realistic radiation and conduction losses.

Improving LOFAR Solar Calibration at Very Low Frequencies: New Methods and a Flare Case Study — Hyoyin Gan

Hyoyin Gan and Christian Vocks

The Leibniz Institute for Astrophysics Potsdam (AIP)

Solar radio imaging is complex because the Sun is an extended, bright, and highly variable source. Rapid solar radio emission—lasting from sub-seconds to seconds—requires high-precision calibration to resolve fine structures. This task is especially challenging at very low frequencies (< 80 MHz), where ionospheric effects introduce significant distortions. The standard LOFAR pipeline typically relies on a two-step process: (1) calibrating against a steady reference source during "quiet" periods and (2) transferring those solutions to active flare data.

Previously, we built a new calibration method incorporating a sun model to improve the calibration. In this talk, we present an updated framework that integrates a 'quiet' solar model with iterative rounds of phase-only self-calibration. We evaluate this method by comparing image quality and signal-to-noise ratios (SNR) against traditional strategies. Finally, we demonstrate the effectiveness of this approach through a case study of solar flares, using our optimised pipeline to process LOFAR observations at very low frequencies (20-80 MHz).

Strong particle acceleration in small flares — Iain Hannah

Hannah, I. G. and Kerr, G. K

University of Glasgow

RHESSI observed 10,000s of microflares (GOES A,B class events) demonstrating particle acceleration and heating like larger events (Christe et al. 2008, Hannah et al. 2008). But one microflare showed unusual X-ray emission > 50 keV and a strong type III radio burst, evidence of strong electron acceleration, even though it was a low A-class microflare (Hannah et al. 2007). Recently dozens of these “hard microflares” have been found with Solar Orbiter on STIX (Battaglia et al. 2024, Saqri et al. 2024), showing similar behaviour of small flares accelerating electrons to high energies. Here we present a reanalysis of the original RHESSI hard microflare, not only in the context of the STIX work, but also using more advanced approaches than the 2007 paper – particularly in terms of the X-ray spectral fitting to determine the properties of this non-thermal energy release. This will then be used to investigate how the solar atmosphere responds to such an energy input through RADYN simulations, helping to understand the energy transport and deposition.

Modelling density profiles along the propagation path of active region jets on August 25, 2011 — Jasmina Magdalenic Zhukov

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Solar coronal jets are narrow and fast propagating ejections of hot plasma, observed at different energies. They are considered to be related to the reconnection between open and closed field lines in the active regions (AR) or at the edges of coronal holes. The AR jets occur regularly at a small-scale, and when they occur on a larger scale, they are related to CMEs and SEPs. Understanding the process of jets requires study of spectral and imaging data across different energy regimes. In this study of AR jets, together with the generally employed multiwavelength analysis, we employ also MHD modelling of the corona in order to determine the coronal plasma density along the propagation path of jets and associated type III radio bursts.

We focused to series of jets observed on August 25, 2011, all originating from the NOAA AR 1271 situated near the west solar limb. We found that studied jets were clearly associated with the three distinctive type III radio burst groups, observed around 11:30, 12:40 and 14:10 UT. The type III radio bursts were observed by ground-based and space-based radio instruments. While Nancay radioheliograph observations allowed us to establish association of type III bursts and jets, direction finding observations by STEREO and WIND spacecraft were employed to estimate the propagation path of radio emission. After estimating type III bursts source positions employing radio triangulation method we modelled the solar corona, with the 3D MHD model of solar wind EUHFORIA, at the obtained source positions.

Our results show that somewhat particular propagation path of type III radio bursts, and possibly also the coronal density profiles (estimated through radio observations and modelling), were probably influenced by the fast solar wind propagating along the open magnetic field lines sourced from the nearby coronal hole.

Probing the Solar Flare Acceleration Region Using Multi-Wavelength Timing and Radio Imaging — Jinge ZHANG

Zhang, J., Reid, H. A. S., James, A. W., Morosan, D., Briand, C., Bhunia, S.

Observatoire de Paris - PSL

Magnetic reconnection in solar flares accelerates electrons upward and downward, producing type III radio bursts and hard X-ray (HXR) emissions. The complex Radio/HXR temporal relationship has led to varied estimates of acceleration region heights and unclear origins of timing delays. We have developed a timing-based framework to estimate the acceleration-region height by interpreting observed delays as arising from differing propagation paths and velocities of electron beams. We combine delay measurements with type III source height and electron speeds from radio and HXR observations, with radio imaging providing independent constraints. Applying this framework to the 11 December 2022 flare using NenuFAR, NRH, and Solar Orbiter/STIX observations, the acceleration region was located in the corona, with consistent results from the timing and imaging approaches. We also found indications of variability between electron injection episodes within the event. This approach offers a new method to localize flare acceleration regions and link timing delays to electron transport.

AstroDoncel Data Centre: New-Generation Space Weather Services from the e-Callisto Network — Joaquín Martínez del Hoyo + Javier Bussons Gordo

Martínez del Hoyo, J.; Bussons Gordo, J.; Prieto Mateo, M.; Borque López-Torres, J.; García Delgado, J.J.

Space Research Group, Universidad de Alcalá (Spain)

Solar Radio Bursts (SRB) are continuously monitored by the e-Callisto network, an efficient Space Weather Instrument comprising more than 80 low-cost radio spectrometers around the world. Near-real-time burst reports including event date, time and spectrograms from the stations reporting positive detection are automatically generated and made public at the AstroDoncel website.

The goal of this new e-Callisto Data Center is to provide services conducive to a substantial reduction of analysis time and effort by the end user and to a more efficient response to potentially disruptive Space Weather episodes. This includes an SRB online catalogue that is traceable, interoperable, re-usable by international research teams, with confidence metrics and links to complementary instruments either ground- or satellite-based missions and geophysical observations.

In this work* we present: samples of the new e-Callisto catalogue, including the so-far-unreported period 2012-19; automatic derivation of additional SRB event parameters associated to burst intensity and type which can play a significant role in future definition of alert levels; full-day spectrograms combining the frequency ranges covered by triggered[^] stations; and a new detection scheme, alternative to AI, based on statistical analysis of spectrogram signals.

*Work funded by Junta de Comunidades de Castilla-La Mancha and the European Union through the CELESTINA-SPARC project (SBPLY/23/180225/000071).

Feature-Based Solar Radio Burst Classification in the e-Callisto Network — Jorge Borque López-Torres

Borque, J.; Bussons Gordo, J.; García Delgado, J.J.; Martínez del Hoyo, J.; Prieto Mateo, M.

Universidad de Alcalá

Solar Radio Bursts (SRBs) are continuously monitored by the e-Callisto network, an efficient space weather instrument comprising more than 80 low-cost radio spectrometers distributed around the world. Near-real-time burst reports, providing event date, time interval, and spectrograms from the stations reporting positive detections, are automatically generated and made publicly available through the AstroDoncel website.

SRBs are classified into different types according to their morphology in the frequency-time domain, reflecting the physical processes occurring in the outer layers of the Sun. In this work, we present a feature-based automatic SRB classification model. By extracting more than 80 features from the spectrograms, we developed a classification framework operating in three modes: a binary SRB/non-SRB mode; a 4-class mode discriminating Type III, Type II, Type IV, and RFI/background; and a complete mode that achieves strong performance across 8 classes.

We compare the results obtained using 15-minute and 1-minute FITS files, based on a dataset containing more than 5,000 SRBs from the e-Callisto network collected between 2024 and 2026, together with selected observations from 2012 to 2019. Preliminary experiments indicate promising classification performance across all three modes, particularly for the reduced and complete schemes. These results suggest that the proposed feature set captures relevant information for SRB characterization. We also present an example of daily classification monitoring within the e-Callisto web, where newly detected SRBs are assigned burst types together with confidence metrics derived from model probabilities and inter-station consistency.

Automated Derivation of SRB Intensity Levels from e-Callisto Dynamic Spectrograms — Juan José García Delgado

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Solar Radio Bursts (SRBs) are continuously monitored by the e-Callisto network, an efficient Space Weather Instrument comprising more than 80 low-cost radio spectrometers around the world. Near-real-time burst reports providing event date, time interval and spectrograms from the stations reporting positive detections are automatically generated and made public at the AstroDoncel website. Expanding this catalogue with physically meaningful and quantifiable parameters is essential for advancing the study of SRBs.

In this work, we present an automated methodology to derive additional physical information from detected events, focusing on SRB localization in the time-frequency domain and intensity classification. Our input consists of spectrogram segments extracted within the temporal windows identified by the detection pipeline. While SRBs ideally exhibit a well-defined intensity maximum, radio frequency interference (RFI) often dominates the signal. To address this, we combine morphological and intensity-based criteria with post-processing steps to distinguish SRBs from RFI, achieving a localization accuracy above 97%. Based on this, we construct a dataset of ~1400 events and develop two models: one based on Y-factor estimates, and another combining Y-factor and signal-to-noise ratio (SNR), enabling probabilistic classification into three intensity levels.

Our results demonstrate that SRBs can be reliably identified even in the presence of strong RFI. The proposed models assign an intensity class to each event as a new catalogue parameter. This framework advances SRB parametrization and enables future developments such as an automated burst delimitation, extraction of frequency bounds and temporal extent, and derivation of properties such as drift rates. Ultimately, this approach improves the characterization of solar radio activity and its potential implications for space weather monitoring.

Comprehensive study of solar type II radio bursts and the properties of the associated shock waves — Karthik Bhandari

Bhandari, K., Morosan, D., Normo, S.

University of Turku

Type II radio bursts are solar radio emissions generated by electrons accelerated by coronal shocks. These bursts are typically found close to expanding coronal mass ejections (CMEs), making them valuable for studying the properties and dynamics of CME-driven shocks in the solar corona.

Here, we aim to determine the regions in the solar corona where shock waves accelerate electrons and determine their characteristic properties. To do this, we combine radio observations of type II solar radio bursts with magneto-hydrodynamic (MHD) simulations of the solar corona.

We analyse ten type II radio bursts from Solar Cycle 25 exhibiting emissions in the 150-300 MHz frequency range. The novelty of this study lies in using radio imaging data for all type II bursts to examine the positions of the radio sources. The radio source positions, combined with a geometrical fitting of the CME shock and the MHD simulations, are used to determine essential shock parameters at the acceleration region, such as the Alfvén Mach number M_A and shock normal angle relative to the magnetic field θ_{BN} . The shock parameters are then combined with the properties of the radio emission and the associated eruption in a comprehensive study.

We found that for all events, the Type II bursts are located near or inside coronal streamers. These streamer regions have a low Alfvén speed while the estimated shock speeds are high, resulting in the formation of super-critical shocks ($4.5 < M_A < 9.8$) at the type II locations. However, in most events, type II bursts are located at oblique shocks rather than near-perpendicular geometries, suggesting that the shock structure is more complex at local scales than the simple spherical shock models usually applied to CME shocks.

Our results suggest that CME-streamer interaction regions are necessary for the generation of type II bursts, as they provide ideal plasma conditions for the formation of super-critical shocks and the subsequent acceleration of electrons.

Geometric constraints on radio triangulation accuracy in multi-spacecraft systems — Ketaki Deshpande

Deshpande, K., Magdalenic, M., Rodriguez, L.

KU Leuven & Royal Observatory of Belgium

Ground-based radio imaging observations are limited to metric wavelengths due to the ionospheric cutoff, restricting studies of radio emissions at larger wavelengths to space-based observations. To overcome this limitation, space-based instruments provide direction-finding (DF) measurements that enable three-dimensional (3D) localization of radio sources employing radio triangulation technique. DF observations from at least two spacecraft need to be combined to estimate the 3D position of radio emission sources.

In this study, we investigate how the accuracy of radio triangulation depends on the relative position of the observing spacecraft and direction of the propagation of radio emission. We analyze 35 type III radio bursts observed simultaneously by at least two spacecraft, i.e., STEREO-A, STEREO-B, and Wind, with different separation angles ranging from 0° to 180° . We studied behaviour of the radio source sizes, defined by the 3D region bounded by the shortest distance between the two wave vectors. We have found the clear U-shaped dependence of the source sizes on the spacecraft separation angle, with the convection point at around 90° . This behavior indicates that optimal triangulation accuracy is achieved for separation angles between 60° and 120° . We also demonstrate that additionally to the spacecraft configuration angle between the burst propagation direction also significantly influences localization accuracy. Moreover, we have found that the radio source sizes have more clear dependence on the relative position of the observing spacecraft and direction of the propagation of radio emission than on the observing frequency.

Additionally, we reconstruct the 3D propagation paths for all 35 events. Our results show that type III radio bursts do not strictly follow Parker spiral magnetic field lines and are not confined to the ecliptic plane, challenging commonly held assumptions about their propagation.

Plasma flow in the solar atmosphere along magnetic fields driven by the Kelvin force — Kiyoto Shibasaki
Shibasaki, L.

Solar Physics Research Inc.

The solar atmosphere is in a plasma state. Magnetic field lines are frozen into the plasma; they move together perpendicular to the field. Along the magnetic field, however, the primary forces governing dynamics are the gas pressure gradient force and gravity. Consequently, in a steady state, hydrostatic equilibrium is achieved. However, observations reveal temperature-dependent upward/downward flows along the magnetic field. In this work, we propose a mechanism to explain flows along the field. Charged particles exhibit gyro motion in a magnetic field due to the Lorentz force, thereby possessing a magnetic moment. However, statistical thermodynamics states that magnetism does not arise in classical physics (Bohr-van Leeuwen Theorem). This theorem can be proven incorrect. Magnetism does arise in classical physics, and plasma possesses a magnetic moment. The Kelvin force acts on the magnetic moment, with a component parallel to the magnetic field. Since plasma is diamagnetic, it experiences a force directed toward regions of weaker magnetic field strength. In the solar atmosphere, the magnetic field is weaker upward, so the magnetic Kelvin force points upward. Moreover, this magnetic moment is proportional to the plasma temperature. When this upward Kelvin force exceeds the downward gravity force, the plasma flows upward. This explains the temperature-dependent upward/downward flows. Introducing the magnetic moment in this way allows us to understand the dynamics of plasma along the magnetic field in the solar atmosphere.

The Deep Learning-Based Dual-Branch Multimodal Fusion Model for Solar Flare Prediction — Limin Zhao

Limin Zhao, Xingyao Chen, Xiaoshuai Zhu, Dong Zhao, Yihua Yan

Beihang University

Solar flares are intense eruptive events caused by the rapid release of magnetic energy, often impacting Earth's space environment through electromagnetic radiation and high-energy particles. Accurate flare prediction is critical for space weather forecasting. However, many existing deep learning approaches often rely on single-modal inputs or shallow feature fusion, limiting their ability to capture complementary information. In this study, we propose a dual-branch multimodal fusion deep learning model for predicting 24-hour solar flares. The model integrates magnetograms and magnetic parameters through cross-attention mechanisms, followed by cross-scale interactions at the feature level to enhance multi-scale representation. It is designed to perform both binary prediction of $\geq$ C-class flares and multi-class classification of C, M, and X-class flares. To ensure rigorous evaluation, we employ a stratified group five-fold cross-validation scheme to preserve class representativeness and adopt a splitting-before-sampling strategy based on NOAA active region numbers to prevent data leakage. Experimental results show that the model achieves a TSS of 0.661 and an HSS of 0.658 for binary $\geq$ C-class prediction, while notably attaining a TSS of 0.780 and an HSS of 0.775 for X-class flares in the multi-class task. Compared with existing approaches, the model demonstrates superior performance in predicting intense X-class flares, effectively suppresses the false alarm rate, and exhibits strong generalization capability.

Proofs of concept in the e-Callisto next generation framework — Manuel Prieto

Manuel Prieto Mateo, Javier Bussons Gordo, Jorge Borque López-Torres, Juan José García Delgado, Joaquín Martínez del Hoyo

University of Alcalá

Solar Radio Bursts (SRB) are continuously monitored by the e-Callisto network, an efficient Space Weather Instrument comprising more than 80 low-cost radio spectrometers around the world. Near-real-time burst reports including event date, time and spectrograms from the stations reporting positive detection are automatically generated and made public at the AstroDoncel website.

In this work, we present the preliminary results of two new concepts under evaluation as part of the initiative for updating e-Callisto. The first concept the “leaky wave” antenna. The Space Research Group of the University of Alcalá developed a new antenna in collaboration with the Technical University of Cartagena. It is a frequency-scanning antenna designed to operate in the 400 MHz to 800 MHz band for solar radio astronomy applications. The antenna was tested and characterized in the laboratory. In this work we present the field tests results for solar radio burst detection.

The second concept is based on a commercial software defined radio device (SDR) as a replacement for the conventional radio receiver/spectrometer. SDRs offer better features for reconfiguration, scalability and in-built digital filtering/processing with respect to the Callisto analog receiver. The SDR was tested in parallel with the conventional e-Callisto RF chain for performance comparison. Preliminary results are presented.

Formation mechanism of short-period quasi-periodic pulsations in radio bursts — Minghui Zhang*Zhang, M.H., Zhang, Y., Yan Y.H., ...*

National Space Science Center, the Chinese Academy of Sciences

To investigate the physical origin and formation mechanism of short-period quasi-periodic pulsations (QPPs) observed in a solar radio burst event, this paper conducts high spatially ($55.6''$ - $10.3''$) and temporally (25 ms) resolution observational research on a radio burst event that occurred on 11 November 2014 using the MUSER radio telescope (with an observation frequency range of 30 MHz to 15 GHz). Through preprocessing of observational data, extraction of pulsation signals, and analysis of spectral characteristics, the core parameter characteristics of QPPs in this event are clarified: the pulsation period is around 1 s, the peak frequency is concentrated in the range of 1-2 GHz, the pulsations show a highly right-hand circular polarization feature and there are four groups of short-period QPPs scattered from 05:06 to 06:31 UT. We also used multiple space instruments from X-ray to extreme ultraviolet to performed the supplementary analysis, including the AIA/SDO, the GOES in 0.1 - 0.8 nm, and the Fermi Gamma-ray Space Telescope. Combined with the magnetic reconnection theory, plasma oscillation mechanism, and nonlinear force-free field (NLFFF) magnetic field extrapolation, the regulatory effects of different formation mechanisms on pulsation period, frequency drift, and amplitude variation are compared and analyzed. The research results show that the formation of QPPs in this event is closely related to the periodic modulation of the small-scale magnetic reconnection process in the source region, such as plasmoid or ultra-thin current sheet. This study provides spectral imaging observational data and theoretical support for an in-depth understanding of the characteristics of the radio fine structures of short-period QPPs in solar eruptions, and provides a valuable reference for the subsequent use of radio observations to diagnose the fine physical conditions of the solar eruption source region.

Solar ALMA observation with very high spatial resolution. — Miroslav Bárta

Barta, M. , Chai, Yi

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Solar ALMA observing mode in its present setup poses severe limitations to extension of ALMA baselines length. Currently, observations in Band 3 (at 100GHz) can be carried out at maximum configuration C-4, Bands 5 and 6 in C-3, and Band 7 in C-2 only. Consequently, the spatial resolution reaches ~ 0.7 arcsec at its best.

Reasons for this constraint lay in our inability to fix the fast phase variations, whose destroying effect increases with the baseline lengths, by the standard calibration procedure. In our recent research, carried out in frame of ESO ALMA Development Study, we have suggested and successfully tested two new approaches for improved observation setup and subsequent calibration, which allow for far longer baselines and, consequently, much better resolution: (1) Time-scale separation between the solar dynamic times and the faster effect of atmospheric turbulence, followed by self-calibration, and (2), extraction of the water-vapor fluctuations directly from the science data in Band 5 and subsequent phase correction in the way used for the non-solar observations. While the latter approach is limited to Band 5 or 7, when we can detect a wing of the water line in the science spectra, the former is universal and only needs the very-high cadence of integrations, currently we use 0.19s (the best in the standard mode so far is 1s). Results: We managed and demonstrated it by real test ALMA observations that the maximum configuration can be extended to C-7 for Band 3, and to C-6 in Band 5, i.e., we have managed to increase the spatial resolution 5x! We will present summary of the procedure and namely its results - movies with unprecedented resolution of 0.15 arcsec obtained by processing of solar test ALMA observations carried out in December 2023.

Why U-Bursts Fail: Kinetic Constraints on Electron Beam Transport — Mohamed Sherif

Sherif, M., Reid, H.

University College London - Mullard Space Science Laboratory

Solar radio U-bursts, generated by near-relativistic electron beams traversing closed magnetic structures, provide diagnostic of plasma dynamics in the solar corona. Despite their shared origin with Type III bursts, U-bursts are observed less frequently. This rarity suggests that electron transport after the apex of closed loops often fails to reach the threshold for coherent emission, implying stringent constraints on the underlying beam-plasma mechanism. This phenomenon remains a theoretical challenge, as it necessitates a detailed understanding of wave-particle interactions in environments of increasing plasma density and varying magnetic convergence. In this study, we use kinetic modelling to examine the evolution of electron beams responsible for U-burst emission. We find that the U-burst morphology is critically sensitive to initial beam parameters; higher beam densities and harder initial velocity distributions yield higher stopping frequencies and beam velocities. Additionally, density inhomogeneities and loop geometry are shown to stifle or facilitate wave growth, serving as a primary factor in the observed variability of U-burst observability and spectral evolution. Our findings provides a physical basis for understanding the comparative rarity of U-bursts and distinguishes them from type III bursts, offering new constraints on electron acceleration and transport in the solar corona.

Spectral component imaging of solar X-ray flares with STIX — Muriel Zoë Stiefel

Stiefel, M.Z., Massa, P., Guidetti, A., Battaglia, M., Krucker, S.

FHNW & ETHZ

Solar hard X-ray (HXR) observations provide key diagnostics of high-temperature plasmas and of nonthermal electron populations present during solar flares and coronal mass ejections. Current HXR instruments rely on indirect methods for imaging, measuring Fourier components, known as ‘visibilities’. In the standard imaging approach, the visibilities of specific energy ranges are extracted from the data and used for image reconstruction. However, the interpretation is often ambiguous, as specific energy ranges can contain overlapping contributions from multiple spectral components, such as a mixture of thermal and nonthermal emission.

We introduce the method of spectral component imaging as developed in Stiefel et al. (2025). In this method, the visibilities of individual spectral components are determined by combining energy-dependent visibilities and results from spectral analysis. We will demonstrate the effectiveness of the method on small B-class flares all the way up to X-class flares and discuss the scientific advantages of the method. Further, we discuss the feasibility of extending this approach to radio imaging techniques, where similar visibility-based methods are employed.

Connecting Electrons at the Sun and in the Heliosphere through X-ray and Radio Diagnostics — Nicole VILMER

Vilmer, N., Paipa, D., Pesini, A., Reid, H., Vecchio, A., Bonnín, X., Maksimovic, M.

LIRA-PSL Paris Observatory CNRS

One of the main objective of the Solar Orbiter mission concerns the production of energetic particles in the heliosphere, and the link between energetic particles at the Sun and in the Heliosphere. For energetic electrons, part of this question can be addressed by combining X-ray and radio observations. The launch of the Solar Orbiter in early 2020 marked a significant milestone, as it is equipped with the capability to simultaneously capture both types of emission. In this contribution, we shall present the results of a first statistical study (Paipa-Leon et al. 2026) based on several tens of events with associated X-ray and interplanetary type III emissions. We will investigate the correlations between the properties of interplanetary type III radio bursts and of their associated HXR X-rays : energy density of the X-ray emitting electrons and characteristics of the radio burst (peak flux, exciter velocity and energy contained in the radio emitting electrons). We will then present preliminary results obtained from the combination of the RPW interplanetary type III catalog recently published (Pesini et al. 2026) and the list of X-ray flares observed by STIX (Hayes et al. 2025) and will investigate how the association rates varies with the distance of the spacecraft to the Sun, the GOES class, the connection with particular active regions, the longitude of the flaring active region.

Reassessing the exciter speed inferred from solar radio bursts — Nicolina Chrysaphi

Chrysaphi, N., Kontar, E. P., Clarkson, D. L.

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Solar eruptive events accelerate energetic particles that can emit intense radio waves throughout the heliosphere. The observed frequency-drift rate of such solar radio bursts is used to obtain the speed with which the exciter of the radio photons is propagating through the heliosphere, whether that is the CME-driven shock or the flare-accelerated electron beam. This exciter speed is often used to infer further values describing fundamental properties of either the local heliospheric environment or the exciter itself. However, relating the observed frequency-drift rate of radio bursts to the exciter speed requires the assumption that the drift rate is purely the result of the exciter's motion. We demonstrate that this is not the case, presenting two additional contributors with often opposing impacts. We examine the balance between the different contributors and identify instances where the observations represent either an underestimation or an overestimation of the intrinsic exciter's motion. We calculate the correction factor between the true and observed values, allowing us to infer the true properties of the exciters of radio emissions, improving the ability of solar radio bursts to remotely diagnose accelerated particles throughout the heliosphere.

Fluctuations in Solar Type III Radio Burst Spectra Observed by Parker Solar Probe and Solar Orbiter — Panisara Thepthong

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LPC2E, CNRS, University of Orléans

Solar type III radio bursts are generated by electron beams accelerated during solar flares and propagating through the heliosphere. Type III radio bursts can be generated by the excitation of Langmuir waves by the electron beam via the bump-on-tail instability, followed by their conversion into electromagnetic emission near the local plasma frequency via nonlinear processes. Alternatively, in an inhomogeneous plasma, the presence of density fluctuations can enable the linear conversion from a Langmuir wave to electromagnetic waves. Observational and theoretical studies have suggested that density fluctuations may contribute to the formation of structured patterns in the radio emission of type III bursts. In this work, we analyze data from Solar Orbiter (SolO) and Parker Solar Probe (PSP) across a wide range of heliocentric distances and observational angles to investigate the spectral structures, or fluctuations, of the type III radio emissions. We first define the maximum spectrum of the bursts by taking the maximum flux in the temporal profile at each frequency, and then investigate both statistically and individually the spectral fluctuations. The statistical analysis of 595 type III events and of their individual fluctuations shows a strong agreement between measurements from the two spacecraft, with no clear dependence on longitude separation, suggesting that local effects are responsible for these fluctuations. Our observations also suggest that there is a continuum of fluctuations between the large fluctuations, which are commonly associated with striae or structured type III bursts, and smaller fluctuations that were not studied before.

3D reconstruction of solar flare X-ray sources from combined STIX/HXI visibilities — Paolo Massa

Massa P., Palumbo B., Stiefel M. Z., Ryan D. F., Collier H., Su Y. , Piana M., Krucker S.

University of Applied Sciences and Arts Northwestern Switzerland (FHNW)

The Spectrometer/Telescope for Imaging X-rays (STIX) on board the ESA Solar Orbiter satellite and the Hard X-ray Imager (HXI) on board the ASO-S mission observe the X-ray emission from solar flares. The two telescopes rely on an indirect imaging technique and measure the intensity of the X-ray radiation that is modulated by means of grid pairs. The data provided by the two instruments can be interpreted as visibilities, i.e., 2D Fourier components of the flaring X-ray sources. However, in the case where STIX and HXI observe the same event from a different vantage point, the datasets of 2D visibilities they provide can be combined to form a sampling of the 3D Fourier transform of the flaring X-ray emission. This combined dataset allows the reconstruction of the 3D morphology of the X-ray sources in solar flares.

We present the results of the 3D reconstruction problem in the case of the X9.1 GOES class flare occurred on October 3, 2024. At that time, the separation angle between STIX, the flaring site and HXI was close to 90 degrees, which is an ideal configuration. We validate our 3D imaging method by comparing the resulting reconstructions with the 2D ones that are independently obtained from STIX and HXI visibilities. Finally, we show how the 3D results can be utilized to estimate altitude and radial velocity of the thermal looptop source as a function of time.

Advancing Solar and Space Weather Science in the LOFAR2.0 and SKAO Era — Pietro Zucca

Pietro Zucca

ASTRON - Netherlands institute for radio astronomy

The Low Frequency Array, LOFAR, is now a key instrument for radio studies of the Sun, the heliosphere and the ionosphere. It offers high time, frequency and spatial resolution, together with a wide field of view. These features make LOFAR very well suited for solar and space weather studies. In this contribution I will present recent results and future plans in the context of the LOFAR2.0 upgrade and the coming Square Kilometre Array Observatory, SKAO.

Current work with LOFAR includes detailed imaging and dynamic spectra of solar radio bursts and coronal structures, and continuous monitoring of ionospheric disturbances. These observations help us understand how energy and particles propagate through the corona, the heliosphere and the near Earth environment, and they provide inputs for models that aim to improve space weather forecasts. The LOFAR IDOLS project, Incremental Development of LOFAR Space weather, operates a dedicated station that observes the Sun and the ionosphere almost continuously. These data are used both for space weather science and for monitoring the ionospheric conditions that affect LOFAR astronomy.

In the LOFAR2.0 era, the array will have higher sensitivity, better stability and more flexible observing modes. It will be possible to observe with the low band and high band antennas at the same time, which will give continuous coverage over a much wider frequency range. This will improve imaging of fine structures in the solar corona and allow better tracking of the full evolution of space weather events from the low corona into interplanetary space.

A key development is the effort on polarization and in particular on circular polarization. New calibration and imaging pipelines now deliver first Stokes V images of solar and planetary radio emission with LOFAR. These first Stokes V results demonstrate that we can begin to measure the magnetic field structure in the corona and in the emission regions of planetary magnetospheres. They also provide a new way to monitor instrumental leakage and to test the stability of the full polarization signal path in LOFAR2.0.

I will also give an overview of the current status of SKAO and of the plans of the Sun heliosphere ionosphere, SHI, science team. SKA Low in Australia is in the middle of its construction phase, with many thousands of antennas already in place and work ongoing on the central processing facility. First test images and first fringes show that the system is progressing toward its first array configuration. SKA Mid in South Africa is also advancing toward early array milestones. The SHI team is preparing science use cases that link current precursors and pathfinders with future SKA studies of the quiet Sun, solar bursts, heliospheric transients and ionospheric structure. Together these facilities will provide multi scale, multi frequency radio coverage of the Sun and the near Earth environment, opening new windows on the physics of solar activity, heliospheric dynamics and ionospheric structure and driving significant advances in fundamental research.

X-ray Spectroscopic Diagnostics of Confined and Eruptive M-Class Solar Flares Using SoLEXS/Aditya-L1 — Prakhar Singh

Singh, P., Pant, V., Joshi, B., Sarwade, A.

Aryabhata Research Institute of Observational Sciences (ARIES)

Solar flares and coronal mass ejections (CMEs) are two major sources of magnetic energy release in the solar atmosphere; however, the physical conditions that distinguish eruptive flares from confined ones are still not fully understood. In this study, we use soft X-ray spectral observations from the Solar Low Energy X-ray Spectrometer (SoLEXS) onboard Aditya-L1 to perform a statistical investigation of the thermal and compositional properties of eruptive (CME-associated) and confined (non-CME) solar flares. We analyze a sample of M-class flares observed during 2024–2025, including both eruptive and confined events. For each flare, we perform time-resolved spectral analysis to derive key plasma parameters: temperature (T), emission measure (EM), and elemental abundances, along with soft X-ray fluence, FIP bias, and flare rise and decay times. These parameters are statistically compared between eruptive and confined flare events to identify differences in thermal evolution, abundance evolution, and energy release characteristics. This study focuses on differences in energy partitioning between eruptive and confined flares. We hypothesize that confined flares exhibit higher peak temperatures, whereas eruptive flares tend to have larger emission measures. Additionally, we find that low-FIP elements (Fe and Ca) in eruptive flares evolve toward more photospheric abundances compared to confined events, which may suggest differences in the origin and transport of flare plasma. Furthermore, we investigate whether differences exist in elemental abundance recovery, soft X-ray fluence, FIP bias, and flare rise and decay times between the two flare categories.

Detection of Linearly Polarized Emissions from Metric Solar Radio Bursts — Puja Majee

Majee, P., Dey, S., Kansabanik, D., Oberoi, D., Mondal, S.

National Centre for Radio Astrophysics, Pune, India

Solar radio bursts are coherent plasma emissions driven by energetic non-thermal electrons accelerated through magnetic reconnection or coronal shocks. The polarimetric properties of these bursts are among the most promising diagnostics for estimating the local coronal magnetic field and provide valuable insights into the underlying emission mechanisms and local plasma conditions. Plasma emission theory predicts these metric solar emissions to be circularly polarized, though propagation effects such as scattering and mode-coupling can significantly reduce the observed degree of circular polarization below the theoretical expectation. Linear polarization, however, has long been considered to be absent, as the conventional wisdom holds that any linearly polarized component would be washed out due to differential Faraday rotation encountered during coronal propagation. As a consequence, any signal appearing in Stokes Q and U has usually been attributed to the instrumental leakage, and reducing them to zero has often been used as a constraint during polarization calibration of solar radio observations. This picture has recently been challenged by the detection of linear polarization in type I and type III bursts from simultaneous observations with the Murchison Widefield Array (MWA) and the upgraded Giant Metrewave Radio Telescope (uGMRT), and, more recently, by the detection of linearly polarized emission also from a type II burst using the MWA. These findings also have broader implications for understanding the emission mechanism of stellar radio bursts, where linearly polarized component are often observed and key insights are drawn from our understanding of analogous solar phenomena. In this talk, I will present the detection and characteristics of linear polarization components of metric solar emissions, focusing on a type II burst, and briefly discuss possible generation mechanisms for the observed linear polarization.

Simulations of SKA Observations for solar emissions — Rohit Sharma

Sharma, R., Nindos, A. and Oberoi, D.

Space, Planetary & Astronomical Sciences & Engineering (SPASE) Indian Institute of Technology Kanpur, India

The Sun is a plasma laboratory. Characteristics of different plasma systems, from the photosphere to the corona, can be captured by studying emissions at radio wavelengths, which originate from a diverse set of emission mechanisms. The spectro-temporal structure of these emissions, their flux densities and polarimetric properties can vary dramatically across different phenomena, magnetic topologies, and emission processes.

Even during periods of low solar activity, solar radio emissions exhibit variability in frequency and time. These emission features can be attributed to magnetic reconnection events crucial for the nanoflare-based coronal and chromospheric heating theories. Recent radio observations have detected emission features with flux densities at milli-Solar Flux Unit (SFU) levels, potentially linked to such small-scale flares. The upcoming Square Kilometre Array Observatory (SKAO) promises increased sensitivity and better imaging quality, which could help further characterise these faint emissions over its precursors.

Mock radio data simulations provide an opportunity to anticipate the possible limits of a given instrument for complex solar radio sources. I will present on solar and heliophysics science cases using mock data simulations along with SKAO observations of faint milli-SFU emission population using Karabo-SKA simulator. Various aspects relating to using optimal data analysis, e.g. deconvolution for SKA-low and SKA-mid, characterisation of self-noise, etc., will be discussed. In addition, the SKAO array assemblies will improve in a phased manner. This mock radio solar data allows us to plan observation strategies and synergies with other solar instruments.

The Solar Radio Playground: Probing Multi-Channel Correlations via Interpretable Diffusion Models for the SKA Era — Roman Farrugia

Farrugia, R., Panos B.,

FHNW I4DS

As we approach the era of the Square Kilometre Array (SKA), the solar physics community faces the challenge of extracting physical insights from petabyte-scale, multimodal radio data. In this work, we develop a "Solar Radio Playground": a synthetic, controllable dataset comprising 8 frequency channels. These data are generated using the FORWARD toolset, driven by magnetic field data from the Solar Dynamics Observatory's HMI instrument, and provided through a collaboration with IIT Kanpur. This setup provides a controllable environment to benchmark sophisticated AI methodologies against known physical ground truths.

While diffusion models represent the state-of-the-art for image generation, their inherent lack of explainability remains a significant hurdle for scientific application. In this work, we demonstrate how these models can be repurposed as powerful explainability tools rather than opaque black boxes. By applying a pixel-wise Mutual Information (MI) decomposition, we generate saliency maps that explain the model's decision-making process during reconstruction. This approach allows for the quantification of exactly which spatial features in a given channel inform the model's prediction of another channel. This technology is currently under incremental development and is being iteratively upgraded in realism to become a flexible, multi-dimensional, and possibly time-lagged framework for probing physical drivers within the SKA data product proper.

Hard X-ray and Radio emissions from the high corona associated with CMEs — Säm Krucker

Säm Krucker

FHNW

Hard X-ray bremsstrahlung emissions are strong diagnostics of electron acceleration during the magnetic energy release process that drives solar flares. Solar Orbiter's unique vantage point away from the Earth-Sun line allows us to use limb-occultation as a diagnostic tool to investigate faint hard X-ray source located above the main flare site which are generally lost in the dynamic range of hard X-ray telescopes. We present new results combining Solar Orbiter/STIX and ASO-S/HXI to study hard X-ray sources associated with CMEs and compare them to radio observations. Our results highlight that high coronal hard X-ray sources are an integral part of solar eruptive events and they are energetically important despite of their intrinsically faint appearance.

Spectropolarimetric analysis of herringbones in a type II solar radio burst — Sanna Normo

Normo, S., Morosan, D. E.

University of Turku

Type II solar radio bursts are well-known signatures of fast electrons accelerated by coronal shock waves. Beyond their classical fundamental-harmonic structure, type II burst can exhibit more complex morphologies, characterised by multiple emission lanes often composed of fine structures, such as herringbones. The presence of these different fine structures reflects the complex and turbulent nature of the shock and the ambient coronal plasma where the observed radio emission is generated. In this study, we aim to study the variation in intensity and polarisation for individual fine structures composing type II bursts. To do this we use high time and frequency resolution NenuFAR dynamic spectra that allow us to extract the intensity-time profiles of individual herringbones in both Stokes I and V and compare their evolution across time and frequency. Here, we report on a complex type II event observed by NenuFAR on 23 September 2025, which exhibits multiple emission lanes and highly circularly polarised herringbones.

Physics-based flare forecasting method with the large Italian Radio Telescopes — Sara Mulas

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INAF - OAC

Solar eruptive events can cause serious hazards to human technologies, therefore a flares forecast system is among the utmost goals of Space Weather. High radio frequency observations with the Italian network of large single-dish radio telescopes resulted in ~ 450 solar images between 2018 and 2023 in K-band frequency range (18–26 GHz).

Solar radio mapping at these frequencies allows the probing of the Active Regions (ARs) chromospheric magnetic field close to the Transition Region, where strong flares and coronal mass ejection events occur. Enhanced magnetic fields up to 2000 G determine anomalous spectra in the ARs brightness compared to pure free-free emission, due to the addition of a steeper gyro-resonance component associated with circular polarisation up to 40%.

In our work we showed that the detection of AR spectral flattening implies a very high probability of a strong flare occurrence within 30 hours (89% statistical precision). With the inclusion of other relevant radio AR parameters, the precision improves up to 97%, proving a high reliability of our physics-based flare prediction method (Mulas et al. 2026 with a national INAF press realise: <https://www.media.inaf.it/2026/01/09/sundish-brillamenti-solari/>). As a future perspective, we are expanded this work to include also the forecast of Coronal Mass Ejections.

Small scale structures within solar noisestorms observed with LOFAR — Soham Dey

Dey S., Oberoi D., Patra D., Zucca P., Mondal S.

National Centre for Radio Astrophysics

Solar noise storms are among the most commonly observed active radio emissions from the Sun at meter wavelengths. The observed angular sizes of radio sources in the solar corona are expected to be dominated by scattering from density inhomogeneities, setting a minimum observable source size. Models of anisotropic radio-wave scattering predict source sizes of few arcminutes at frequencies around 140 MHz.

We present interferometric imaging observations of a solar noise storm at 140-190 MHz using the LOw Frequency ARray (LOFAR), including Dutch remote baselines extending to ~ 75 km. This is the first high angular resolution imaging study of noise storm structures at these frequencies with LOFAR. We detect small-scale structures in the noise storm emission with angular sizes significantly smaller than those expected from standard scattering models, while a type-III burst observed during the same period shows source sizes consistent with predictions. We show that the noise storm sources are spatially coincident with coronal loop-top regions visible in extreme ultraviolet observations. Drawing on recent work by Mondal et al. 2025, we propose that the over-dense coronal loop environment in which noise storms originate can substantially reduce the effective scattering path length, thereby explaining the anomalously compact source sizes. These results demonstrate that noise storms and type-III bursts experience markedly different propagation environments.

Solar jets as the direct source of impulsive solar energetic electrons — Song Tan

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The precise solar origin and acceleration mechanism of impulsive solar energetic electron (SEE) events remain a subject of debate. While these events are commonly associated with flares and type III radio bursts, previous statistical studies comparing in situ electron release times with Hard X-ray (HXR) peaks have reported weak correlations and unexplained time delays, hindering a clear understanding of particle injection and transport. In this study, we present a statistical analysis of over 100 impulsive SEE events detected by EPD on board Solar Orbiter between 2020 and 2023 (as a part of the Comprehensive Solar Energetic Electron event Catalogue (CoSEE-Cat)). Using high-cadence EUV imaging from SDO/AIA, we pinpoint the onset times of associated EUV jets and compare them with solar release times (SRTs) derived via Velocity Dispersion Analysis (VDA). Our results reveal a significantly improved correlation between VDA-derived SRTs and jet onset times compared to traditional HXR- or type III-based proxies. We further examine the temporal relationship between EUV jets and their associated type III radio bursts to explore whether jet-driven reconnection provides the open magnetic field pathways along which energetic electrons escape into the heliosphere. These findings support EUV jets as the primary driver of impulsive SEE injection and highlight the importance of combining EUV jet observations with radio diagnostics for accurate timing analysis.

Imaging spectroscopy reveals spike-like repeating solar burst pairs in the solar corona — Suli Ma

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Solar radio bursts exhibit complex fine structures that reveal intricate coronal plasma dynamics. Here, we report a new type of spike-like repeating burst pairs, characterized by two short-lived (0.1-2s), narrowband components separated by ~ 4 s at frequencies 30-50 MHz. Using high-resolution dynamic spectra and spectroscopic imaging, we analyzed 613 burst pairs, measuring their durations, bandwidths, drift rates, flux densities, and spatial characteristics. Imaging links sources to an active region, with earlier components spatially concentrated and delayed components displaced and exhibiting reduced drift rates. Radio-wave propagation simulations support the delayed bursts as turbulent echoes of harmonic emission in anisotropic coronal plasma.

The location of the burst sources high in the corona suggests ongoing magnetic reconnection and electron acceleration well above typical flare heights. Our findings offer new insights into coronal turbulence effects while advancing diagnostics of coronal plasma and the elusive nature of solar radio echoes from ground-based transmitters.

SPOTLIGHT on decimetric millisecond bursts: Insights from uGMRT solar observation — Surajit Mondal

Mondal, S., Dey, S., Patra, D., Majee, P. , Oberoi, D., Kansabanik, D., and the SPOTLIGHT collaboration

National Centre for Radio Astrophysics

Millisecond radio bursts in the decimetric bands, while discovered back in 1977, have been investigated in only a handful of studies till date. The primary reason behind this is that very few instruments have the ability to record data at the time resolution required for characterizing these bursts. Among these few studies, there are even fewer instances where it was possible to image the corresponding radio source. Even in those instances, the imaging was possible only with data integrated over time spans much larger than the burst duration, raising questions about the true nature of the millisecond radio source. In this investigation, we, for the first time, image a millisecond radio burst at a time resolution of 1.25ms and frequency resolution of 198kHz, and thus resolve the radio burst both in spectro-temporal as well as spatial domains. This ultra-high time and frequency resolution imaging capability has been enabled by the Survey for sPporadic radiO burstsTs via a commensaL multIbeam Gpu-powered Hpc at the gmrT (SPOTLIGHT) backend operating at the Upgraded Giant Metrewave Radio Telescope (uGMRT). We use these data to show that even at these time-frequency resolutions, the observed emission is incredibly fragmented in the spectro-temporal plane, with very rapid intensity fluctuations. While a frequency-drift is visible in the dynamic spectrum, the radio source locations show rapid movements, with no apparent systematic drift across frequency. We explain this scenario by hypothesising that the emission is highly affected by scattering, which is responsible for rapid shifts of the apparent source location. Since such rapid shifts are likely possible when the emission frequency is close to the plasma frequency, these observations can be used to constrain the emission mechanism responsible for the generation of the millisecond radio burst.

Three-dimensional mapping of coronal magnetic field and charged particles in an M-class solar flare —

Tatiana Kaltman

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Probing physical conditions in solar flare is crucial for understanding coronal energy release and particle acceleration.

Simultaneous X-ray observations from Solar Orbiter/STIX and Hinode/XRT, taken from different vantage points, allowed stereoscopic reconstruction of the flaring loop. We use imaging spectroscopy data of the SOL2021-05-07 flare available from Expanded Owens Valley Solar Array to derive maps of the magnetic field strength, thermal and nonthermal electron densities, and the power-law index of nonthermal electrons via gyrosynchrotron modeling. By correlating positions of microwave and thermal X-ray sources, we associate the reconstructed 3D coordinates of the X-ray source with the microwave-derived plasma parameters.

In addition to the derived 3D maps of the magnetic field strength, we also compute 3D maps of the Alfvén speed and plasma beta in a flaring volume, revealing a magnetically dominated environment. These spatially resolved diagnostics provide valuable constraints for models of magnetic reconnection and flare dynamics, representing a step toward realistic 3D characterization of energy release in solar eruptive events.

The Nature of Electron Density Inhomogeneities in the Middle and Outer Corona — Timothy Bastian

T. S. Bastian

National Radio Astronomy Observatory

The VLA was used to observe background radio sources during solar minimum in 2019 to transilluminate the foreground solar corona. A total of 18 sources was observed during the course of a solar rotation to observe along 63 lines of sight. These probed the corona from 1.9-10.5 solar radii for a wide range of heliolatitudes. Density inhomogeneities in the corona lead to angular broadening of the background source. For the outer corona ($r > 5-6$ solar radii), the (anisotropic) spectrum of electron density inhomogeneities is well fit by a power law with a Gaussian cutoff. Interpreted in terms of a turbulent spectrum, the spectral index, inner scale, and degree of anisotropy are deduced. These results are largely consistent with past work although there are several new and interesting wrinkles -- for example, we find that the inner scale is isotropic. For the middle corona, defined as the range 1.5-6 solar radio, the situation is less clear. I suggest that “deterministic scattering”, rather than scattering on turbulent inhomogeneities, may dominate here.

A New Radio-Quiet e-CALLISTO Site in the Azores: Extending Solar Radio Burst Coverage across the North Atlantic — Valente Cuambe

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Solar radio bursts are a key diagnostic of flare and CME-driven particle acceleration and represent a major source of interference for radio-based technological systems, including Global Navigation Satellite Systems (GNSS). Continuous, low-latency monitoring of solar radio emission is therefore essential for both scientific understanding and operational space-weather services. The global e-CALLISTO network provides near real-time dynamic spectra at metric and decametric wavelengths, but its performance critically depends on local radio-frequency interference (RFI) conditions.

Within the SOLAR-Az project, a comprehensive RFI survey was conducted across Santa Maria and Flores islands (Azores, Portugal) to identify suitable locations for the deployment of new e-CALLISTO spectrometers. Broadband measurements covering 10~MHz--1.7~GHz were carried out at more than twenty candidate sites, and site suitability was quantified using statistical indicators of peak power, mean levels, and variability, ensuring safe operation under low-noise amplification without receiver saturation.

The survey identifies a single location at Farol da Maia (Santa Maria Island) as a genuine radio-quiet zone. In particular, the area near the abandoned whale factory exhibits no detectable interference from commercial, military, or emergency communication systems across the native CALLISTO band, enabling full-band observations without additional filtering. This site is therefore selected for the installation of the e-CALLISTO antennas, representing the cleanest radio environment identified in the Azores, among the sites investigated.

On EIT Waves Relationship with Coronal Mass Ejections, Solar Flare, Coronal Holes, and Solar Wind Phenomena — VIRENDRA VERMA

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In the present paper, we studied relationship of EIT waves with Coronal Mass Ejections (CMEs), solar flares, coronal holes (CHs), and solar winds. Here, we have studied the relationship between the Extreme Ultraviolet Imaging Telescope (EIT) waves phenomena with solar flares, coronal holes, solar winds, and CMEs events. The EIT/ SOHO instrument recorded 176 EIT events during the above period (March 25, 1997-June 17, 1998) and the EIT waves list was published by [1] T. After temporal matching of EIT wave events with CME phenomena, we find that corresponding to 58 EIT wave events, no CME events were recorded and thus we excluded 58 EIT wave events from the present study. Out of 176 EIT wave events, only 106 are accompanied by CMEs. The correlation study of the speed of EIT wave events and CMEs events of 106 events shows poor correlation $r=0.32$, indicating that the EIT waves and CMEs events do not have a common mechanism of origin, and also indicate that some other factor is working in the formation of CMEs from EIT waves. Further, We have also matched the spatial matching EIT wave sources as indicated by [1] with CHs and flares and found that CMEs appear to be associated with EIT wave phenomena and CHs. Earlier [2], and [3] indicated that the CMEs may have been produced by some mechanism, in which the mass ejected by solar flares or active prominences, gets connected with the open magnetic lines of CHs (source of high-speed solar wind streams) and moves along them to appear as CMEs. Most recently [4] proposed a methodology to understand the origin of CMEs through magnetic reconnection of CHs open magnetic field and solar flares.

In the present paper, we proposed a scenario/ 2-dimensional model, in which the origin of CMEs through reconnection of EIT waves and solar winds coming from the CHs and also found that the calculated CMEs velocity after reconnection of EIT waves and solar winds coming from the CHs are in very close to the observed CMEs linear velocity. We also calculated the value of the correlation coefficient between the observed linear velocity of CME events and the calculated value of CME velocity after reconnection and found the value as $r=0.884$. The value of correlation as $r=0.884$ is excellent and supports the proposed methodology. Further study is in progress to understand the role of solar winds in origin of CMEs. Finally, we have also discussed the role of solar wind phenomena in the formation of CMEs phenomena with other solar phenomena, in the latest scenario of solar heliophysics phenomena. Finally, we have also discussed the role of solar wind phenomena in the formation of CMEs phenomena with other solar phenomena, in the latest scenario of solar heliophysics phenomena.

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Metric Radio Bursts and Their Implications for Solar Activity Observed by MUSER-L (30-400 MHz) — Xingyao Chen

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The Mingantu Spectral Radioheliograph low-frequency array (MUSER-L), operating in the 30-400 MHz range, has been conducting routine solar observations within the 0-8 UT time window since 2025, providing high-cadence imaging spectroscopy with 0.1 s temporal and 1 MHz spectral resolution. During its initial observing phase, MUSER-L has accumulated over 150 days of observations and detected a variety of coherent radio bursts, including type III burst storms, type II bursts, quasi-periodic pulsations, spike bursts, etc. The combination of imaging and dynamic spectral observations offers the opportunity to investigate the spatial and temporal evolution of processes such as electron acceleration, shock propagation, and wave-particle interactions in the solar corona. We present representative events together with an overview of the current data products. As the data processing system is still under development, observations are presently based on a subset of antennas and short baselines, with longer baselines and data processing pipelines still being commissioned. These preliminary results illustrate the potential for combining imaging and spectral diagnostics in the metric regime and provide a basis for future quantitative studies and multi-wavelength investigations of solar eruptive phenomena.

High-resolution Solar Observation with ALMA — Yi Chai

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High spatial resolution in ALMA observations has been one of the major promises for the astronomy and astrophysics communities, and solar researchers are no exception. While solar observations have benefited from unprecedented millimeter/submillimeter data that were unavailable before ALMA, they remain constrained by limited choices of antenna configurations for given bands. So far, these limits on array configurations have not accounted for the possibility of correcting sequences of distorted images using techniques such as self-calibration. Given that higher spatial-resolution ALMA observations have the potential to address key questions—such as the well-known coronal heating problem—we proposed this study to push the maximum baseline length for solar observations as far as possible; the proposal was later accepted as an ESO project. We successfully developed routines to simulate ALMA solar observations using long-baseline array configurations with atmospheric disturbances represented by phase errors, and we compared the self-calibrated results with the unself-calibrated ones. The simulations show that, with the help of self-calibration, it is possible to extend Band 3 observations to configuration C-7 and Band 5 to C-6. A test observation using these setups was carried out, and the data were successfully reduced using self-calibration.

MUSER Observations of Solar Radio Burst Events — Yihua Yan

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Wide-band solar radio observations, spanning decameter to centimeter wavelengths, constitute a key component of the Chinese Meridian Project (CMP) for tracking and monitoring solar eruptions from the Sun's atmosphere into interplanetary space. The Mingantu Spectral Radioheliograph (MUSER) with three arrays at low (30–400 MHz), intermediate (400 MHz–2 GHz), and high (2–15 GHz) frequency bands images the solar atmosphere in 3D from the top-chromosphere up into the mid-corona (Yan et al. 2021, 2025). The MUSER observations have been presented previously (e.g., Zhang et al., 2021). With the newly established calibration systems for MUSER under CMP-II and the new calibration method, further progress on solar radio imaging spectroscopy has been made. Various solar radio bursts have been recorded by MUSER. Data are transmitted to the CMP data center daily. Here we present the recent progress on MUSER for solar and space weather observations.

First Detailed MeerKAT Imaging Spectroscopy of a Solar Flare — Yingjie Luo

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Radio observations provide powerful diagnostics of energy release, particle acceleration, and transport processes in solar flares. However, despite recent advances in radio interferometric imaging spectroscopy, current instruments remain limited by image fidelity and resolution, restricting detailed spectroscopic studies of flaring regions. In this report, we present high-fidelity imaging spectroscopy of an M1.3-class flare observed by the MeerKAT array, a precursor to the future-generation facility SKA-Mid. Emission at these frequencies originates primarily in the low corona, offering direct insight into magnetic reconnection and primary energy-release sites. The observations achieve an unprecedented dynamic range exceeding 10^3 , enabling the simultaneous analysis of intense coherent bursts and much fainter incoherent emission from the active region. Multiple spatially distinct coherent sources are identified, suggesting contributions from different populations of accelerated electrons. In addition, the incoherent emission extends beyond structures seen by AIA, highlighting MeerKAT's capability to detect dilute but hot plasma not accessible to extreme-ultraviolet diagnostics. Combined with contemporaneous hard X-ray observations and magnetic field extrapolations, the radio sources are found to be associated with distinct magnetic structures, providing further insight into the origin and evolution of energetic electrons. These results demonstrate the importance of high-fidelity imaging spectroscopy in disentangling complex flare emission and establish MeerKAT as a powerful tool for solar flare studies, paving the way for future observations with SKA-Mid.

